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Winner of the 2025 Fischer Black Prize

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Tyler Muir is the Donnalisa '86 and Bill Barnum Endowed Term Chair in Management at UCLA Anderson School of Management. Before joining UCLA, Tyler was on the faculty of the Yale School of Management. He earned his Ph.D. in 2013 from the Kellogg School of Management at Northwestern University, where I was fortunate to have served on his dissertation committee.

Tyler has established himself as one of the most insightful and impactful empirical financial economists of his generation. He has made contributions to three areas of research: (i) the impact of intermediaries on asset prices and the macroeconomy; (ii) volatility and risk/return trade-offs; and (iii) the impact of central bank actions on asset markets. I highlight his main contributions in each of these areas.

Tyler's dissertation and the work early in his career belong to the first strand of work. Coming out of the 2008 financial crisis, there was considerable interest in understanding the role of financial intermediaries in driving asset prices and macroeconomic outcomes. "Intermediary asset pricing" was motivated by the failings of the traditional household consumption-based approach to asset pricing. Although theoretical models of intermediary asset pricing have been developed, these models have not been taken to the data systematically. Tyler does this in two papers. "Financial Intermediaries and the Cross-Section of Asset Returns," with Tobias Adrian and Erko Etula and published in *The Journal of Finance* (2014) is the seminal paper in empirical intermediary asset pricing. The paper constructs an intermediary stochastic discount factor from broker/dealer balance sheet data. Intermediary asset pricing models predict that a high marginal utility state for the intermediary is one where leverage is high/intermediary capital is low. The paper constructs this leverage factor in the data and shows that the factor prices many asset returns that traditional measures, such as aggregate consumption, struggle with. The paper, which received the Smith-Breeden Prize in 2011, has been the template for a growing body of empirical work in intermediary asset pricing.

In "Do intermediaries matter for aggregate asset prices?" (*Journal of Finance*, 2021), Tyler Muir and Valentin Haddad take a different approach, deriving and verifying a key prediction of the intermediary asset pricing model: shocks to intermediaries will most strongly impact the prices of assets that are most intermediated. The paper develops an approach for measuring the "most intermediated" dimension and evaluating the intermediary asset pricing model in this dimension. The paper won the DFA first prize in 2021.

Theory indicates that the impact of financial intermediary frictions will be most apparent in financial crises. However, in the modern data sets commonly

examined in the asset pricing literature, there is typically only one crisis, the 2008 global financial crisis. “Financial crises and risk premia” (*Quarterly Journal of Economics*) and “How credit cycles across a financial crisis” (*Journal of Finance*) examine data spanning 150 years, many countries, and many crisis episodes. The *Quarterly Journal of Economics* paper shows that if one examines the consumption disasters considered in the rare-disasters literature, all of the action in asset prices and risk premia occurs in banking-crisis disasters rather than in war disasters. Thus, for understanding asset prices, a researcher can more narrowly focus on financial-crisis disasters. Moreover, the evidence indicates that the state of the financial sector, not fluctuations in aggregate consumption, is the key driver of asset prices and risk premia in these events, underscoring the importance of financial frictions in macroeconomic disasters.

The *Journal of Finance* paper, coauthored with me, examines this connection in the reverse direction, asking how asset price fluctuations shed light on the mechanisms driving banking crises. In models of crises driven by distress in the financial sector, a severe crisis is one where a large negative shock hits a leveraged financial sector. Consistent with this prediction, the paper shows that the macroeconomic impact of a financial crisis is well-explained by the interaction of the jump in credit spreads in a crisis, capturing the size of the negative shock, and pre-crisis credit growth, capturing leverage in the financial sector. Another factor in many narratives of banking crises is a pre-crisis frothy phase. The paper examines this using credit spreads to evaluate the extent of asset market froth and shows that crises are preceded by periods of low credit spreads and high credit growth. Tyler and I then show that a signal constructed from both credit spreads and credit growth has more predictive power for the occurrence of a banking crisis than either credit spreads or credit growth.

Tyler has been a key contributor to the growing body of research on the role of financial intermediation in asset pricing and macroeconomics. For asset pricing, Tyler’s empirical work in testing intermediary asset pricing theories has been enormously influential. For macroeconomics, his innovative use of large-scale historical data sets—linking macro variables with asset prices—has provided valuable insights into the dynamics and aftermath of financial crises.

Tyler’s second strand of research revisits the classical asset pricing question of how risk premia adjust as volatility changes. In a series of three papers beginning with “Volatility Managed Portfolios” (*Journal of Finance*), Tyler Muir and Alan Moreira (2017) show that scaling equity exposure inversely with recent volatility generates consistently higher Sharpe ratios than static strategies; that is an investor can benefit from a timing strategy whereby the investor reduces equity exposure when volatility spikes. The result is surprising given the economics in most equilibrium models: mean-variance preferences strongly imply that Sharpe ratios should rise with volatility. Tyler Muir and Alan Moreira (2019) dig further into this in their paper “Should long-term investors time volatility?” published in *the Journal of Financial*

Economics. Perhaps a resolution of the puzzle can come from the hedging demands of long-term investors. The paper considers this possibility in more depth using a quantitative portfolio choice framework with a long horizon Epstein-Zin investor. However, they show that, quantitatively, given the properties of stock returns over the past century, volatility timing is still beneficial to long-term investors. The third paper in this line of research, “Volatility expectations and returns,” written by Lochstoer and Muir (2022) and published in *The Journal of Finance*, offers a resolution of the puzzle. Tyler and Lars present a model where beliefs are updated too slowly as in models of sticky expectations. They show that the slow updating of beliefs can generate the volatility/return relation in line with the data as well as a profitable volatility timing strategy for an investor who updates their beliefs following Bayes rule.

Tyler’s third strand of research, written with Valentin Haddad and Alan Moreira, considers the impact of central banks on asset prices. Former Federal Reserve Chairman Ben Bernanke famously quipped, “The problem with QE is it works in practice, but it doesn’t work in theory.” In response to this challenge, Tyler, along with Valentin and Alan, has produced research that sheds light on the specific mechanisms through which quantitative easing (QE) influences asset markets, helping to bridge the gap between theory and observed outcomes.

The first paper in this line of work is “When selling becomes viral: Disruptions in debt markets in the COVID-19 crisis and the Fed’s response,” by Haddad, Moreira, and Muir (2021), published in the *Review of Financial Studies*. The paper is an event-study analysis of the Fed’s actions during the COVID crisis and is one of the go-to reads on understanding the Fed’s actions during COVID. The most significant paper in this line of work is their *American Economic Review* (2025) paper “Whatever it takes? The impact of conditional policy promises.” Central bank asset purchases under QE can theoretically work through at least two channels: the central bank purchases directly shift today’s demand curve for the purchased asset, and, the central bank indirectly shifts today’s demand curve by shaping investor’s expectations over how the central bank may buy assets in a future state. Tyler, Valentin, and Alan show that this second “conditional policy” channel is present in the data and is moreover the dominant channel in many cases of QE. While earlier research had indicated the possibility of this channel, their paper is the first to present a methodology to isolate the channel using option prices. Across many central bank purchase events involving the Fed, Bank of Japan, Bank of England, and European Central Bank, they show that a large part of the response to these announcements comes from shifts in the tail states. The conditional policy channel is the dominant channel in the Fed’s announced purchase of corporate bonds in March 2020, where almost no bonds were purchased, but credit spreads reacted very strongly. Their research shows that the market reaction, as revealed via option prices, was entirely due to anticipating a Fed backstop on corporate bond prices.

Haddad, Moreira, and Muir (2024) have continued to work in this impactful line of research around QE policy rules in their working paper “Asset purchase

rules: How QE transformed the bond market". It takes the state-contingency point that the *American Economic Review* paper makes and applies it to thinking about the Treasury and mortgage-backed securities purchases of the Fed in the 2008 financial crisis and recovery. Through a mix of high- and low-frequency data, interpreted through a model, the paper shows how a researcher can extract the QE policy rule that market participants believe. The paper finds that 10-year Treasury yields were about 1.5% lower because of these policies.

There is a large literature on the 2008 and beyond QE policies of central banks. Most of this work constitutes small steps forward. The research that Tyler has done with Valentin and Alan are *giant* steps forward, making both economically substantive and quantitatively important points regarding state-contingency that is shaping the way that researchers understand the impact of central banks on asset prices.

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